NetTrace

Professional Network Traffic Analysis & Security Investigation Using Wireshark

Prepared for: Cybersecurity / SOC Portfolio Submission

Capture analyzed: capture1.pcapng

Analysis platform: Wireshark

Scope: Authorized home-LAN traffic captured by the analyst

Executive Summary

This report documents a structured network-traffic investigation performed with Wireshark against an authorized home-LAN packet capture. The analysis covered protocol distribution, endpoints and conversations, DNS activity, HTTP traffic, TLS/QUIC encryption, TCP reliability indicators, I/O traffic patterns, and Wireshark Expert Information.

The capture contained 132,793 packets. TCP and UDP dominated the traffic, with substantial QUIC and TLS activity. A notable security observation was the discovery of a router administration page served over HTTP at [redacted]`/admin/login.asp`. The capture did not include the corresponding login POST, so the analysis does not claim that router credentials were transmitted in plaintext.

The investigation also identified 1,860 TCP retransmissions (~1.4%), 167 TCP resets, 107 missing DNS responses, and 4,213 QUIC handshake-decryption warnings. These events were treated as investigative indicators rather than evidence of compromise. Overall, the capture demonstrates practical PCAP-analysis skills and shows how to distinguish observable network behavior from unsupported security conclusions.

1. Objectives and Scope

Objectives

• Establish the protocol profile of the capture.
• Identify important local and external communication endpoints.
• Investigate DNS, HTTP, TLS and QUIC activity.
• Assess TCP reliability indicators such as retransmissions and resets.
• Use Wireshark I/O Graphs and Expert Information to support evidence-based conclusions.
• Document security-relevant observations without over-classifying normal network behavior.

Scope and authorization

The analysis is limited to traffic captured from the analyst's authorized home-LAN environment. No attempt was made to access third-party systems, bypass authentication, or obtain credentials from other users. The report describes only what is supported by the captured evidence.

2. Capture Overview

Metric	Value
Capture file	capture1.pcapng
Total packets	132,793
Primary host observed	[redacted]
Local router observed	[redacted]
HTTP packets	190
DNS packets	1,616
TCP packets	88,257
UDP packets	44,095
QUIC packets	42,119
TLS packets	29,559
TCP retransmissions	1,860 (~1.4%)

3. Protocol Hierarchy Analysis

The Protocol Hierarchy view shows that the capture was dominated by TCP and UDP. QUIC represented a large proportion of the traffic and operated over UDP, while TLS accounted for a substantial amount of encrypted application traffic. The presence of HTTP traffic provided a useful contrast because selected HTTP content remained directly readable.

Protocol / Layer	Packets	Share	Analytical significance
TCP	88,257	66.5%	Primary transport protocol; used for connection and reliability analysis.
UDP	44,095	33.2%	Significant datagram traffic; includes QUIC and local discovery.
QUIC	42,119	31.7%	Substantial encrypted UDP-based web/transport activity.
TLS	29,559	22.3%	Encrypted application traffic; payload is not directly readable.
DNS	1,616	1.2%	Domain-resolution metadata visible in the capture.
HTTP	190	0.1%	Plain HTTP traffic; includes a router administration page.

Protocol / Layer	Packets	Share	Analytical significance
ARP	70	0.1%	Local LAN address-resolution activity.

Analytical note: percentages are the values displayed by Wireshark in the supplied Protocol Hierarchy view; they are not intended to imply that every protocol row is mutually exclusive at every layer.

4. Endpoint and Conversation Analysis

Endpoint and conversation analysis was used to identify the principal participants and high-volume connections. The workstation 192.168.1.10 generated substantial encrypted connections to external services, predominantly over TCP/443. The capture also contained DNS communication with Cloudflare public resolvers.

Selected conversation evidence

Observed flow	Port	Packets / context	Assessment
[redacted] ↔ 163.70.144.60	5222	2,369 total packets in conversation	Observed application/service traffic; no malicious conclusion.
[redacted] ↔ 3.166.96.16	443	7 total packets	Encrypted HTTPS/TLS communication.
[redacted] ↔ 18.97.36.17	443	414 total packets	Encrypted HTTPS/TLS communication.
[redacted] ↔ 23.193.165.42	443	76 total packets	Encrypted HTTPS/TLS communication.
[redacted] → 23.193.165.43	80	13 total packets	Plain HTTP flow; warrants application-level review.

The selected conversations are representative evidence extracted from the supplied Wireshark conversation data. They are not presented as threat verdicts.

5. DNS Analysis

DNS traffic provided useful visibility into service discovery and application intent. The workstation 192.168.1.10 was observed querying Cloudflare public DNS resolvers at 1.1.1.1 and 1.0.0.1. Visible queries included `www.canva.com`, `telemetry.canva.com`, `chrome.cloudflare-dns.com`, and `www.whatsapp.com`.

Local multicast DNS traffic was also observed from a LAN device involving `_googlecast._tcp.local`. This is consistent with local service discovery. The observed DNS names were not independently classified as malicious.

Security relevance: DNS remains valuable during encrypted-traffic investigations because domain-resolution metadata can reveal which services a host attempts to reach even when the subsequent application payload is encrypted.

6. HTTP Investigation — Router Administration Interface

A particularly useful finding was an HTTP request from the workstation to the local router:

```
GET /admin/login.asp HTTP/1.1
Host: [redacted]
```

The router returned `HTTP/1.0 200 OK` and identified the server as `Boa/0.93.15`. The returned HTML contained a login form with username and password fields and a form action of `/boaform/admin/formLogin`. The page also referenced Realtek-specific content and JavaScript responsible for processing the login form.

Security assessment: the management interface is demonstrably being delivered over HTTP, which provides no transport encryption. However, the capture did not contain the actual POST request generated when the login form is submitted. Therefore, plaintext credential exposure is **not confirmed** by this dataset.

7. TLS and QUIC Analysis

The capture contained 29,559 TLS packets and 42,119 QUIC packets. QUIC traffic appeared as non-readable encrypted/binary payload when inspected without session secrets. This provides a practical demonstration of the difference between readable HTTP application data and encrypted modern web/application traffic.

Wireshark Expert Information reported 4,213 QUIC 'Failed to decrypt handshake' events. These warnings are consistent with the analyst not possessing the cryptographic session secrets required to decrypt the observed QUIC sessions. They should not be treated as evidence of malicious activity.

8. TCP Reliability Analysis

The display filter `tcp.analysis.retransmission` identified 1,860 packets, approximately 1.4% of the complete 132,793-packet capture. The retransmissions were distributed across multiple TCP conversations, with many visible conversations using port 443.

Possible explanations include packet loss, congestion, wireless interference, receiver limitations, or other transient network conditions. Retransmissions alone are insufficient to classify the traffic as malicious.

9. I/O Graph Analysis

The I/O Graph used one-second intervals and showed a predominantly low-rate traffic baseline interrupted by short-lived bursts of higher packet activity. A noticeable increase in activity was visible near the end of the capture. The graph supports temporal traffic analysis but does not, by itself, identify the cause of a spike.

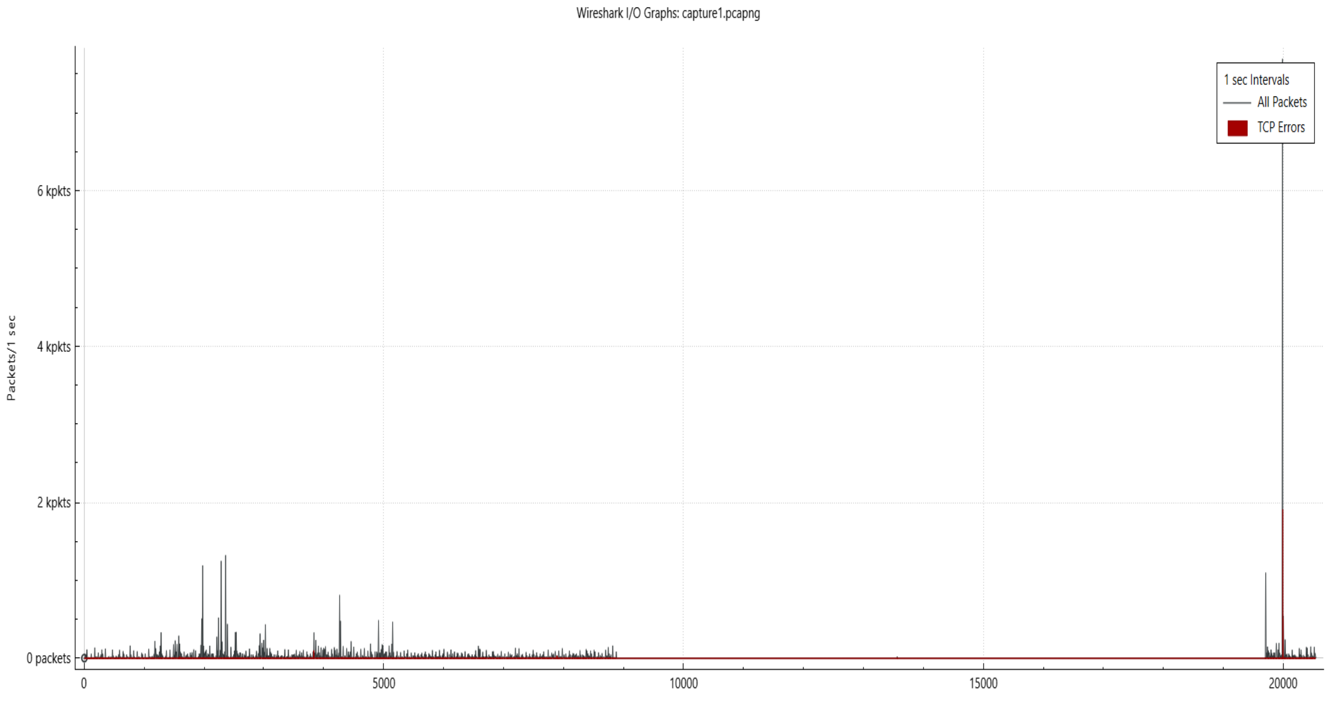
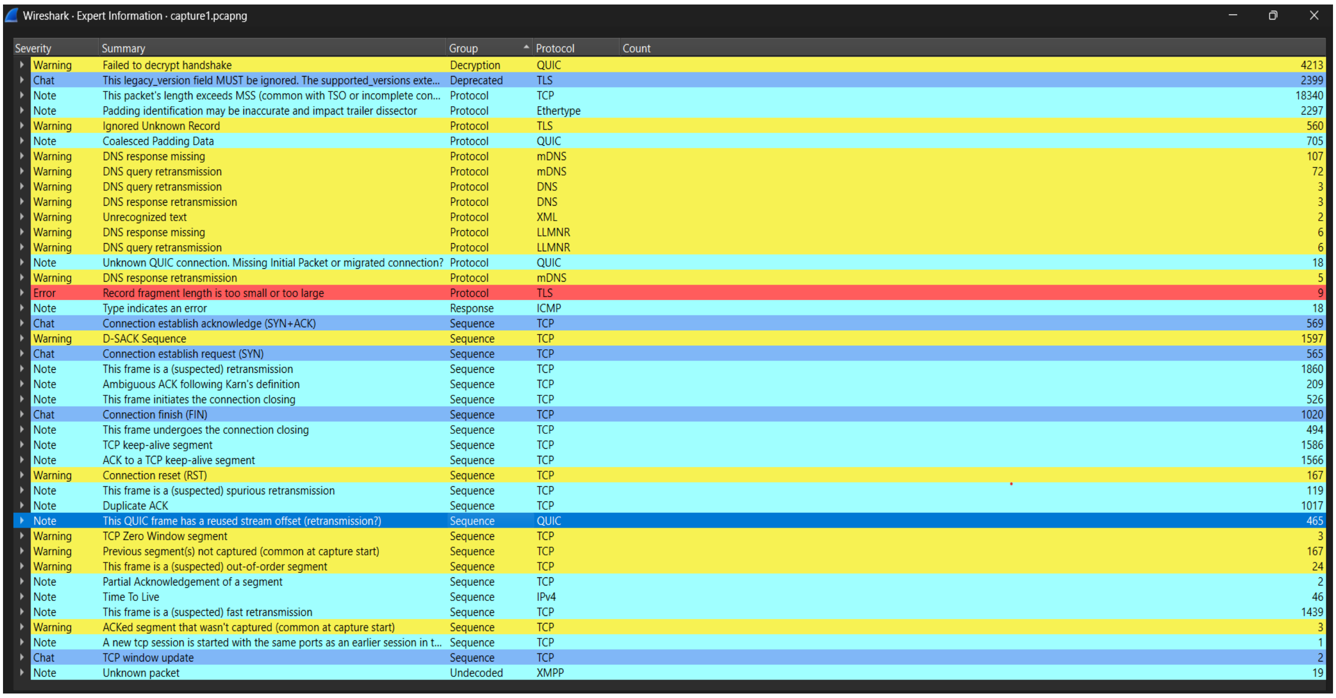


Figure 1. Wireshark I/O Graph for the analyzed capture.

10. Expert Information Review

Wireshark Expert Information provided a consolidated view of protocol and sequence observations. The most important entries are summarized below. Expert Information is treated as an investigative aid rather than an automatic threat detector.

Expert observation	Count	Interpretation
QUIC handshake decryption failed	4,213	Encrypted traffic could not be decrypted without the required secrets; not proof of an attack.
TCP retransmission	1,860	Matches the dedicated retransmission filter; possible network-quality indicator.
TCP D-SACK sequence	1,597	TCP acknowledgement/recovery behavior.
TCP keep-alive	1,586	Normal long-lived connection behavior.
TCP FIN	1,020	Normal connection termination events.
Duplicate ACK	1,017	TCP acknowledgement behavior; can accompany loss/reordering.
QUIC coalesced padding	705	QUIC protocol behavior.
TCP RST	167	Connection resets; requires context.
DNS response missing	107	Some DNS queries had no corresponding response in the capture.
TCP zero-window	3	Receiver temporarily advertised no available receive window.
Out-of-order TCP segment	24	Some TCP segments were observed out of sequence.


Wireshark · Expert Information · capture1.pcapng

Severity	Summary	Group	Protocol	Count
Warning	Failed to decrypt handshake	Decryption	QUIC	4213
Chat	This legacy_version field MUST be ignored. The supported_versions exte...	Deprecated	TLS	2399
Note	This packet's length exceeds MSS (common with TSO or incomplete con...	Protocol	TCP	18340
Note	Padding identification may be inaccurate and impact trailer dissector	Protocol	Ethertype	2297
Warning	Ignored Unknown Record	Protocol	TLS	560
Note	Coalesced Padding Data	Protocol	QUIC	705
Warning	DNS response missing	Protocol	mDNS	107
Warning	DNS query retransmission	Protocol	mDNS	72
Warning	DNS query retransmission	Protocol	DNS	3
Warning	DNS response retransmission	Protocol	DNS	3
Warning	Unrecognized text	Protocol	XML	2
Warning	DNS response missing	Protocol	LLMNR	6
Warning	DNS query retransmission	Protocol	LLMNR	6
Note	Unknown QUIC connection. Missing Initial Packet or migrated connection?	Protocol	QUIC	18
Warning	DNS response retransmission	Protocol	mDNS	5
Error	Record fragment length is too small or too large	Protocol	TLS	9
Note	Type indicates an error	Response	ICMP	18
Chat	Connection establish acknowledge (SYN+ACK)	Sequence	TCP	569
Warning	D-SACK Sequence	Sequence	TCP	1597
Chat	Connection establish request (SYN)	Sequence	TCP	565
Note	This frame is a (suspected) retransmission	Sequence	TCP	1860
Note	Ambiguous ACK following Karn's definition	Sequence	TCP	209
Note	This frame initiates the connection closing	Sequence	TCP	526
Chat	Connection finish (FIN)	Sequence	TCP	1020
Note	This frame undergoes the connection closing	Sequence	TCP	494
Note	TCP keep-alive segment	Sequence	TCP	1586
Note	ACK to a TCP keep-alive segment	Sequence	TCP	1566
Warning	Connection reset (RST)	Sequence	TCP	167
Note	This frame is a (suspected) spurious retransmission	Sequence	TCP	119
Note	Duplicate ACK	Sequence	TCP	1017
Note	This QUIC frame has a reused stream offset (retransmission?)	Sequence	QUIC	465
Warning	TCP Zero Window segment	Sequence	TCP	3
Warning	Previous segment(s) not captured (common at capture start)	Sequence	TCP	167
Warning	This frame is a (suspected) out-of-order segment	Sequence	TCP	24
Note	Partial Acknowledgement of a segment	Sequence	TCP	2
Note	Time To Live	Sequence	IPv4	46
Note	This frame is a (suspected) fast retransmission	Sequence	TCP	1439
Warning	ACKed segment that wasn't captured (common at capture start)	Sequence	TCP	3
Note	A new tcp session is started with the same ports as an earlier session in t...	Sequence	TCP	1
Chat	TCP window update	Sequence	TCP	2
Note	Unknown packet	Undecoded	XMPP	19

Figure 2. Wireshark Expert Information summary from the capture.

11. Security Findings and Risk Assessment

ID	Finding	Severity	Evidence-based assessment
F-01	Plain HTTP Router Administration Interface Observed	Medium	The host 192.168.1.10 requested /admin/login.asp from 192.168.1.1 over HTTP. The response was HTT
F-02	Significant Encrypted QUIC/TLS Activity	Informational	The capture contained 42,119 QUIC packets (31.7%) and 29,559 TLS packets (22.3%). The inspected Q
F-03	DNS Activity Reveals Service Metadata	Informational	The workstation 192.168.1.10 generated DNS queries to 1.1.1.1 and 1.0.0.1. Observed names included
F-04	TCP Retransmissions Observed	Low / Operational	1,860 packets were identified with tcp.analysis.retransmission, approximately 1.4% of the 132,793-packe
F-05	TCP Resets and DNS Response Gaps	Low / Investigative	Wireshark Expert Information reported 167 TCP connection resets and 107 missing DNS responses. Th

12. Recommendations

• Prefer HTTPS for router and device administration whenever the device supports it; avoid exposing administrative interfaces over plaintext HTTP.

• Restrict router management access to trusted LAN hosts and disable remote administration unless it is explicitly required.

• Keep router firmware current and replace unsupported/obsolete firmware or management software where practical.

• For future captures, preserve synchronized timestamps and capture on the relevant interface so complete connection flows can be reconstructed.

• For deeper QUIC/TLS analysis, collect session secrets in an authorized test environment; do not attempt to decrypt third-party traffic without authorization.

• Investigate repeated retransmissions, resets, or missing DNS responses only when they correlate with a specific host, service, or incident timeframe.

13. Methodology and Reproducibility

The investigation followed a layered workflow: (1) preserve the original PCAP, (2) establish protocol hierarchy, (3) review endpoints and conversations, (4) isolate DNS and HTTP, (5) inspect TLS/QUIC behavior, (6) assess TCP reliability indicators, (7) visualize traffic over time, and (8) review Expert Information. Display filters were used to reduce the packet set to specific analytical questions.

Purpose	Wireshark display filter
All HTTP	http
HTTP requests	http.request
HTTP POST	http.request.method == "POST"
TCP retransmissions	tcp.analysis.retransmission
Initial TCP SYN	tcp.flags.syn == 1 && tcp.flags.ack == 0
DNS queries	dns.flags.response == 0
QUIC	quic
TLS	tls
Port 80	tcp.port == 80

14. Limitations

This report is based solely on the supplied packet capture and the Wireshark views derived from it. The capture does not establish the identity, ownership, or intent of every external endpoint. Encrypted traffic cannot be interpreted at the application-payload level without the appropriate keys/session secrets. The missing router login POST prevents confirmation of whether credentials were transmitted in plaintext. Wireshark Expert Information can contain events caused by normal network conditions or capture artifacts and therefore requires contextual validation.

15. Conclusion

The NetTrace investigation demonstrates a practical Wireshark workflow for examining real network traffic. The capture contained a mixture of local discovery, DNS, HTTP, TCP, TLS, and QUIC traffic. The most significant security observation was the presence of a router administration page delivered over plaintext HTTP, while the majority of application traffic observed was encrypted. TCP retransmissions, resets, DNS gaps, and QUIC decryption warnings were documented as investigation indicators rather than incorrectly classified as attacks.

Overall assessment: No confirmed compromise or malicious activity was established from this capture alone. The strongest actionable observation is the use of HTTP for the local router administration interface, which should be reviewed and hardened where supported.

Appendix A — Evidence Checklist

✓ Protocol Hierarchy captured
✓ Endpoint/Conversation analysis performed
✓ DNS queries reviewed
✓ HTTP router administration page identified
✓ TLS/QUIC traffic reviewed
✓ TCP retransmissions quantified
✓ I/O Graph reviewed
✓ Expert Information reviewed
✓ Security conclusions limited to observed evidence